

MEDICAL CORE · PROTOTYPE

rTMS Navigation Prototype

Rapidly turn tracking and coordinate data into a repeatable Slicer navigation prototype.

EVIDENCE STATE**Prototype · Ongoing****PERIOD****2024.07 - present****PORTFOLIO TIER****Medical Core****PRIMARY CAPABILITY****Medical Navigation & Visualization**

Connected Slicer structure, tracking devices, coordinate flow, and UI into a repeatable research-navigation environment.

Problem and system boundary

01 / SLICER ARCHITECTURE

Slicer architecture

Connected device input, transforms, targets, and UI into a repeatable structure.

PROBLEM

Tracking data and medical-image targets needed a repeatable flow that did not hide coordinate assumptions.

Connected Slicer structure, tracking devices, coordinate flow, and UI into a repeatable research-navigation environment.

PUBLIC BOUNDARY

This is a research prototype screen and does not claim clinical efficacy, quantitative accuracy, or commercial deployment.

Domain experts review the tracking hardware and research workflow.

EVIDENCE STATE

Prototype · Ongoing / NeuroPilot navigation module: tracker, patient-marker and coil-marker status, distance and tilt guidance to the target, and the aiming guide (project path hidden).

Owned decisions and implementation

MY ROLE

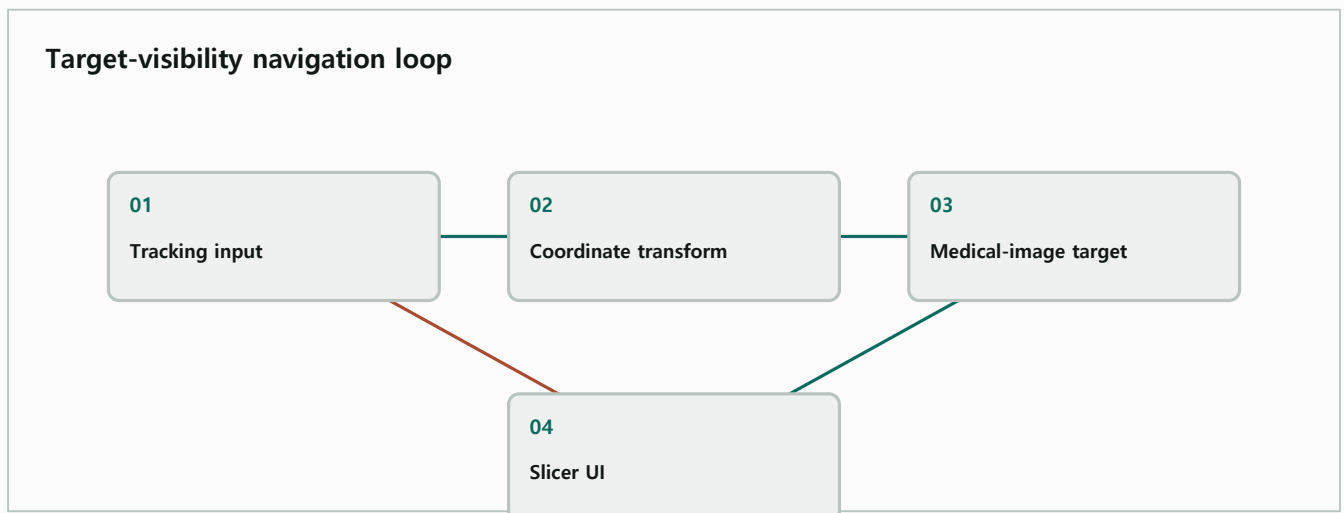
Led the Slicer structure, device integration, coordinate flow, UI, and repeatable experiment environment.

02 / COORDINATE VISIBILITY

Coordinate visibility

Kept coordinate frames and transforms explicit in the prototype.

SYSTEM RELATIONSHIP



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

3D Slicer / VTK / Qt / OpenIGTLink / Optical tracking / C++ / Python

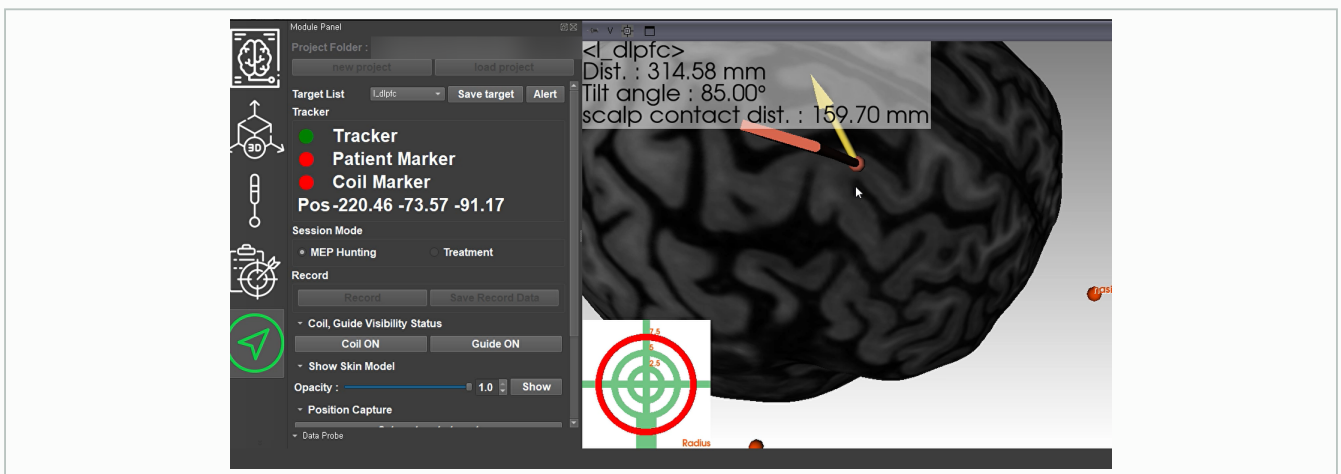
Verification and evidence ledger

03 / REPEATABLE PROTOTYPE

Repeatable prototype

Use repeated execution of the same device, coordinate, and UI flow as evidence.

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

IMAGE / PUBLICLY INSPECTABLE

rtms-navigation-lead-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

rtms-navigation-gallery-01

Approved derivative; caption in portfolio data.

VERIFICATION AND EVIDENCE LEDGER

A repeatable Slicer prototype and visible tracking, coordinate, and visualization flow provide the evidence.

Role, team result, and limits

04 / PROTOTYPE BOUNDARY

Prototype boundary

Clinical efficacy and quantitative accuracy require separate validation.

MY ROLE

Led Slicer structure, device integration, coordinate flow, and UI.

Led the Slicer structure, device integration, coordinate flow, UI, and repeatable experiment environment.

TEAM RESULT

The team shared a working research prototype; this is not extended into a clinical-efficacy or quantitative-accuracy result.

A repeatable Slicer prototype and visible tracking, coordinate, and visualization flow provide the evidence.

LIMITATIONS

This is a research prototype screen and does not claim clinical efficacy, quantitative accuracy, or commercial deployment.

NeuroPilot navigation module: tracker, patient-marker and coil-marker status, distance and tilt guidance to the target, and the aiming guide (project path hidden).

COLLABORATION BOUNDARY

Domain experts review the tracking hardware and research workflow.

Connected Slicer structure, tracking devices, coordinate flow, and UI into a repeatable research-navigation environment.

Recap and joint-development invitation

Rapidly turn tracking and coordinate data into a repeatable Slicer navigation prototype.

PRIMARY CAPABILITY

Medical Navigation & Visualization / 3D Geometry & Registration

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Domain experts review the tracking hardware and research workflow.

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